ANALYSIS · 11 OCTOBER 2026

From a scraper to topic pages you can flip through

Options for developing the News Scraper Platform further, tested on your own library, with costs and a recommended order.

Prepared for Jacob by Claude. The tests ran on a copy of your items and cost nothing; no paid request was made with your API key.

1. In short

You want a page per topic that you flip through the way you flip through videos. Each page says what happened, lists every related source (articles, papers, Reddit, videos), and shows whether the topic is growing or fading, over months and over days.

The platform already collects, searches, translates, and writes a briefing when you ask. Two things are missing. It cannot tell which items belong together by what they are about, only by the words in them. And its topic pages do not keep themselves up to date.

I recommend building in this order:

1. **Give every item an embedding.** An embedding is a list of numbers that stands for what a text is about; two texts on the same subject get lists that lie close together. A free model on your own computer makes them. Everything below depends on this.
2. **Group items into stories**: the same event or the same report, seen in several places.
3. **Rebuild the topic page** so that it updates itself: what is new, the long run, a timeline, the sources by kind, a trend line, and an "up next" list to flip to the next topic.
4. **Add YouTube channels and podcasts** as sources, shown on the topic pages.
5. **Add a trends page**: what is spiking this week and what has been building for months.

Adding more sources can go on alongside all of this.

7 seconds
to give all 380 of your items an embedding with a free model, on your processor

28 groups
found in your library without any AI cost; most are real subjects, three are not

$4–7 a month
estimated Claude Haiku cost at 100 sources and 300 new items a day

$0.32
estimated one-off cost to give all 380 stored items an English summary with Haiku

Two things to do today, without building anything. None of your five sources has an update schedule, so the library only grows when you press the button. Trend lines need steady collection, so set an interval on each source. And switch on *AI summary and tags* for all sources: the test showed that grouping works much better on a clean English summary than on raw page text (section 4.1).

2. Where the platform stands

What you want	What exists now	What is missing
Collect from many sources	Feeds, listing pages, sitemaps, Reddit, built scrapers. 5 sources and 380 items today.	Adding sources in bulk; a schedule (all five update only by hand); YouTube; podcasts.
Find what belongs to a subject	Word search. A topic is two lists of words.	Search by meaning; grouping. A German text that never uses the English word is not found.
A page per topic	The matching items, plus a briefing that Claude writes when you press the button. You have two topics (Anduril, LNG); neither has a briefing yet.	A page that updates itself, a timeline, sources by kind, videos, flipping from one topic to the next.
Trends, long and short	Nothing.	Counts over time, and enough history to count.
Pick out what is interesting	Nothing.	A score per story.

3. The target: one page per topic

The aim is a page per topic that keeps itself current, and a quick way to move from one topic to the next. To get there, every item goes through five stages. The platform has the first and part of the last; the three in blue are what to build.

1. Collect
Scrape sources on a schedule. Add YouTube, podcasts, more feeds.

NEW
2. Understand
English summary and tags (exists, optional), plus an embedding per item.

NEW
3. Group
Items into stories; stories into topics you follow.

NEW
4. Write
Claude writes a headline per story and keeps each topic page current.

5. Show
Topic pages, flipping, a trends page. The reader and cards exist.

Figure 1 is a sketch of the page I would build towards, for your own example. The headlines, dates, counts and the curve are real: they come from your library and from a free worldwide news index (section 4.5). The grey bars stand for text that Claude would write. The home page becomes a wall of topic cards, as YouTube's is a wall of videos: each card shows the topic's newest headline, a small trend line and how many stories are new since your last visit.

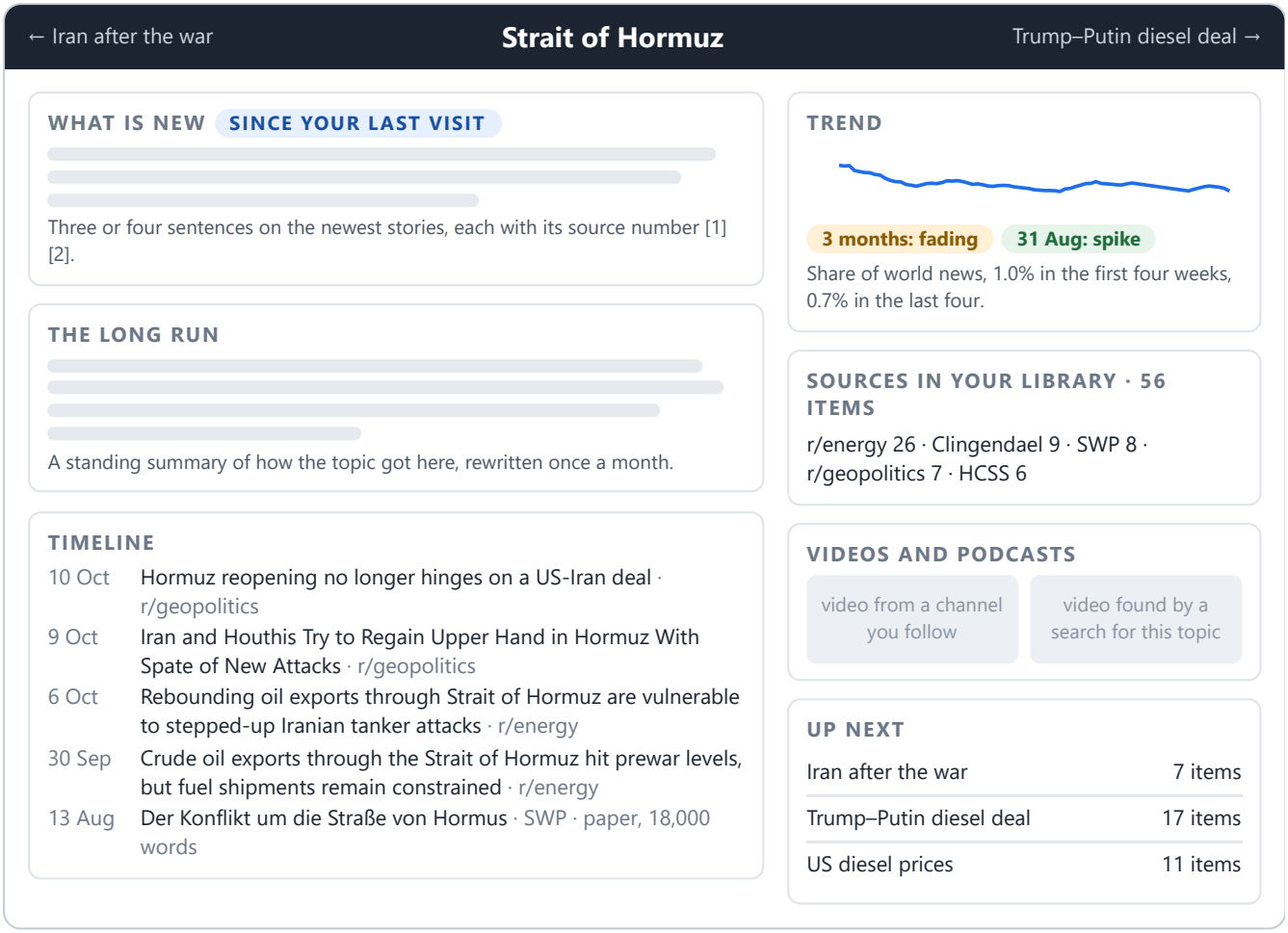


Figure 1. A sketch, not a screenshot. The arrows in the top bar and the “up next” list are what makes it flip like videos: one key press moves to the next topic.

4. The building blocks, and what the test showed

4.1 Embeddings: knowing what an item is about

I copied your 380 items to a temporary folder and gave each one an embedding with a free multilingual model (`paraphrase-multilingual-MiniLM-L12-v2` , a 240 MB download) that runs on your processor. Then I tried searching and grouping with them.

What I measured	Result
Embedding all 380 items	7 seconds, 54 items a second. 300 new items a day would take about 6 seconds.
A larger, stronger model (`multilingual-e5-large` , 2.1 GB)	94 seconds, 4 items a second. Its groups were a little tighter, not clearly better.
One search across 100,000 embeddings	5 milliseconds, in 154 MB of memory. No separate database is needed; SQLite and numpy are enough.
Search “blockade of the Strait of Hormuz”	Returned the Hormuz items, and also two German SWP papers on endangered sea lanes and on land-based anti-ship weapons. The word search for “Hormuz” does not find those two.

What I measured	Result
Search "Is Russia going to attack NATO?"	First result: the Dutch HCSS piece "Poetin is all-in – is een Russische aanval op de NAVO een kwestie van tijd?"

What you get from this step alone: a search box that understands a question in plain English and finds Dutch and German items too, a "more like this" list under every item, and topics that match by meaning instead of by a word list.

> **The most useful finding.** When I fed the model the opening of each stored text, it grouped items by the wrong things: HCSS papers by their copyright page, Clingendael papers by their disclaimer, Reddit items by the line "Shared in r/…", and Dutch items simply because they are Dutch. With only the title and the site's short summary the groups became much cleaner, but Dutch and German items still pulled together by language. A short English summary of every item, which the platform can already write with *AI summary and tags*, should remove most of that. I did not test this last step, because it would have spent your money: about $0.32 for all 380 items on Claude Haiku 5.5, by the platform's own estimate.

> **My pick.** The small free model, fed with the title plus the English AI summary. Store which model made each embedding, so that a better one can replace it later: redoing 380 items takes seconds and 100,000 items about half an hour. The alternative is a paid service, Voyage AI, which Anthropic's documentation points to: $0.02 per million tokens with the first 200 million free, so it would also cost you nothing for years. It is likely better, but it needs another account and key.

4.2 Stories: the same event in several places

A story is a set of items that are almost the same in meaning and appeared within about ten days of each other. Finding them needs no AI: each new item is compared with the stories of the last days and either joins one or starts a new one.

In your library this found **25 stories covering 62 items**. They turned out to be three kinds, all useful:

- **One event, many reports.** The Trump–Putin diesel deal: six items from r/geopolitics and r/energy between 9 and 11 October. Ethiopia: a Clingendael paper of 2 October and an article shared on Reddit on 7 October.
- **One report, two forms.** An HCSS paper and its own announcement; the German and the English edition of an SWP paper.
- **Reposts.** "World's tallest wind turbine" was shared three times in r/energy in one week.

The home page can then show one card per story instead of six near-identical cards, and the number of separate sources covering a story becomes the first measure of how important it is. Only 5 of the 25 stories span more than one source today, because you have five sources and two of them are Reddit. With more news sources this share rises quickly.

4.3 Topics: the ones you make, and the ones the platform suggests

Grouping all 380 items by subject, again without AI, gave **28 groups holding 214 items**. I named them myself from their titles; in the platform Claude Haiku would name them for a fraction of a cent. Figure 2 shows your library laid out so that items on the same subject sit together, and the table lists twelve of the groups.

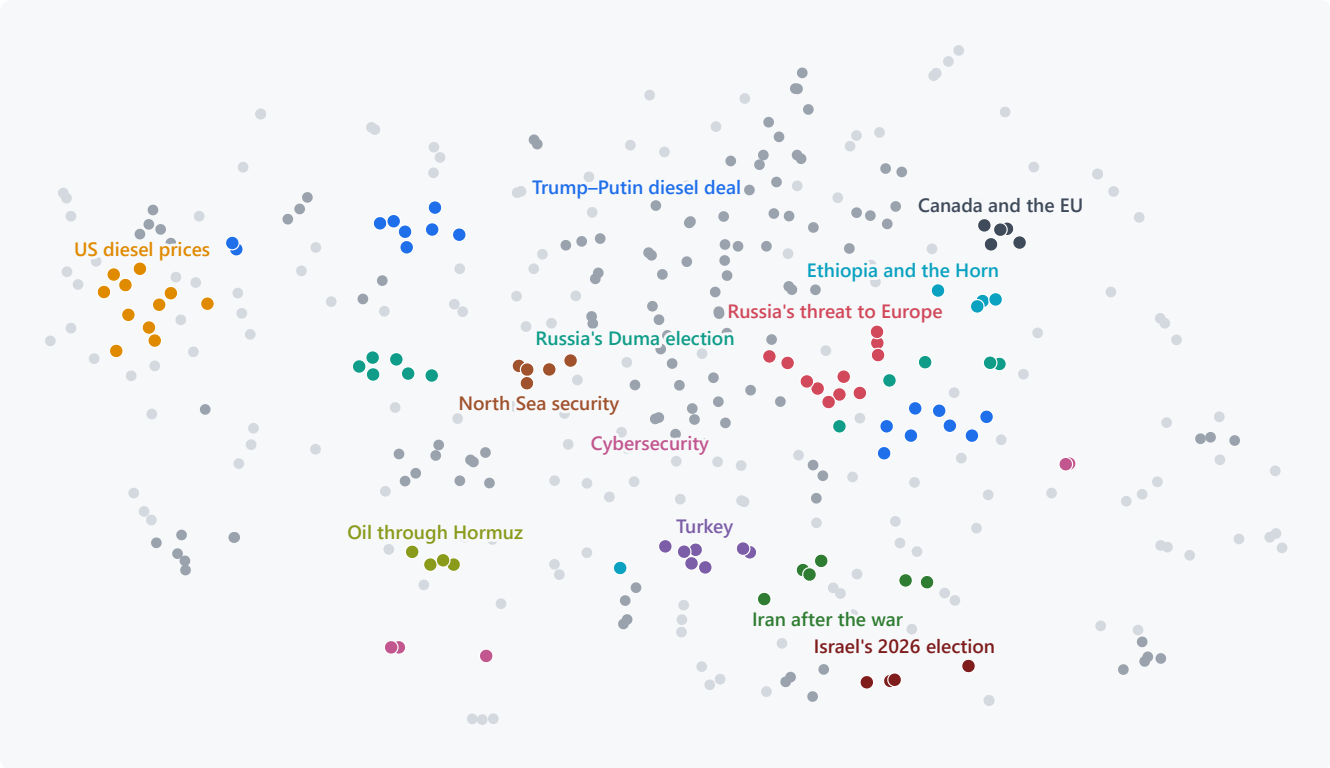


Figure 2. Your library as a map. Each dot is one item; nearby dots are about the same subject. Coloured dots belong to the twelve groups in the table below, dark grey dots to the other sixteen groups, light grey dots to no group.

Group	Items	Sources	Languages	The item closest to its centre
Trump–Putin diesel deal	17	r/geopolitics 9, r/energy 8	English	U.S. official: Trump cut deal with Putin after Zelensky ignored his requests
Russia's threat to Europe	11	r/geopolitics 7, HCSS 2, r/energy 2	English, Dutch	Tim Sweijs | Poetin is all-in – is een Russische aanval op de NAVO een kwestie van tijd?
Russia's Duma election	11	SWP 9, r/geopolitics 2	German, English	Noch eine Scheinwahl in Russland
US diesel prices	11	r/energy 11	English	The U.S. Has a Diesel Shortage. Why Is Red-Dyed Diesel Suddenly on the Table?
Turkey	7	SWP 6, Clingendael 1	German, English	Türkische Außenpolitik nach dem Nato-Gipfel in Ankara
Iran after the war	7	SWP 3, Clingendael 2, r/geopolitics 1, r/energy 1	English	Why the Gulf Monarchies Are Not Breaking Away from the US Despite the Iran War
Cybersecurity	6	SWP 5, HCSS 1	German, English	Multilaterale Kooperation und Vertrauensbildung im Cyberbereich
Ethiopia and the Horn	5	r/geopolitics 2, Clingendael 1, HCSS 1, SWP 1	English	Ethiopia's widening war threatens the Horn of Africa
North Sea security	5	HCSS 4, Clingendael 1	Dutch, English	Twee agenda's, één zee: Hoe veiligheid en natuur op de Noordzee elkaar kunnen versterken

Group	Items	Sources	Languages	The item closest to its centre
Canada and the EU	5	SWP 4, HCSS 1	German, English, Dutch	Kanadas Angebot an Europa: Mehr als nur Handelspartner
Oil through Hormuz	4	r/energy 3, r/geopolitics 1	English	Crude oil exports through the Strait of Hormuz hit prewar levels, but fuel shipments remain constrained
Israel's 2026 election	4	Clingendael 2, r/geopolitics 1, SWP 1	English	Israel's 2026 elections

Not every group is a subject. Three of the larger ones show the problem from section 4.1:

Group	Items	Sources	What went wrong
Mixed bag (Dutch and German texts)	26	HCSS 10, SWP 9, Clingendael 6, r/geopolitics 1	A loose mix of climate, raw materials and EU policy. What these items share is mostly that they are in Dutch or German.
Dutch opinion surveys	13	Clingendael 6, HCSS 6, SWP 1	Dutch opinion surveys and columns, again held together by language.
HCSS house news	10	HCSS 10	"HCSS in the media" round-ups and a vacancy: grouped because they are announcements, not because of a subject.

And 166 items fell into no group of four or more. That is normal for a library of this size: a subject needs several items before it shows up as a group.

> **My pick.** Keep topics as something you decide, and use the automatic groups as suggestions: "17 items about a Trump–Putin diesel deal. Follow this as a topic?" A topic you follow is then defined by a short description and a few example items instead of a word list, and you correct it with two buttons, *belongs here* and *not this*. The word lists you have now can stay as an extra filter for names such as Anduril.

4.4 Topic pages that keep themselves up to date

Today a briefing is written from scratch each time: Claude Opus reads the 30 best-matching items. I estimate that at 10 to 20 cents per briefing; you have not run one yet, so this is not measured. Doing that daily for twenty topics would cost $60 to $120 a month.

The cheaper design keeps a memory per topic and only reads what is new:

- **What is new** is written when new stories arrive, from the standing summary plus the summaries of the new stories only. That is about 3,000 tokens in and 400 out: about $0.0005 on Haiku, 1 cent on Sonnet, 2 cents on Opus.
- **The long run** is rewritten once a month from the monthly notes, not from all items. About 15 to 20 cents per topic on Opus.
- **Every sentence keeps its source number**, as the briefing does now, so you can always check a claim against the item it came from.

The page also shows things that need no AI at all: the timeline of stories, the sources split by kind (news, papers, Reddit discussion, video), the trend line, and the related topics for "up next", which are simply the topics nearest in meaning.

4.5 Trends: the long run and the short run

A trend is a count over time. For each topic the platform counts its share of all new items per week. Two comparisons then give the two views you asked for:

- **Short run:** the last 7 days against the 4 weeks before. Much higher means *spiking*.
- **Long run:** the last 3 months against the 3 months before. This gives *building*, *steady* or *fading*.

Figure 3 shows both for your example, using GDELT, a free index of world news that anyone can query. I asked it how much of each day's news mentioned “Strait of Hormuz”.

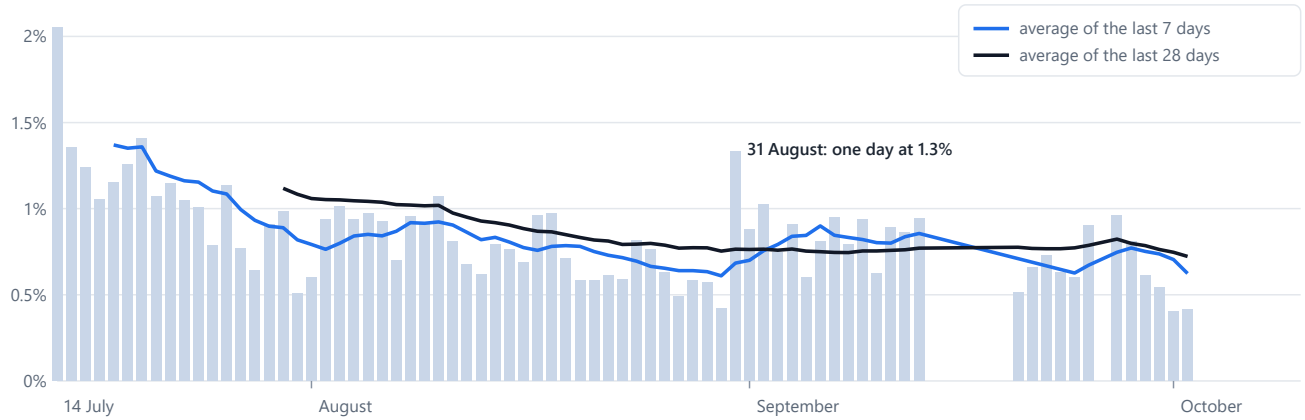


Figure 3. Share of world news articles mentioning “Strait of Hormuz” per day, 14 July to 2 October 2026 (bars), with a 7-day and a 28-day average. The 28-day line shows the long run: from 1.0% in the first four weeks to 0.7% in the last four. The 7-day line catches the short spike at the end of August. Source: GDELT DOC 2.0 API, queried 11 October 2026; days for which GDELT returned almost no data are left out: a stretch in mid-September and the most recent week.

> **Two limits to know before building this.** First, a trend line from your own library shows what you collected, not what happened. Of your 380 items, 238 are dated in the last two weeks, because that is when you scraped and because Reddit feeds only reach back a few days. That is why the schedule matters, and why counting shares instead of raw numbers matters: otherwise every new source makes every topic “rise”. Second, your history starts around September 2026. Long-run lines need either time, or deep scrapes of think-tank archives (which go back years), or an outside index such as GDELT for the world's curve.

4.6 Picking out what is interesting

I would score stories, not single items, and start with signals that cost nothing:

- **Coverage:** how many separate sources carry the story.
- **Weight of the source:** a number you set per source, so that a Clingendael report counts for more than a Reddit post.
- **Newness:** how far the story is from everything already in the library. This surfaces subjects you have not seen before.
- **Attention:** Reddit comments, YouTube views against the channel's usual number.
- **Your own behaviour:** what you open, star or hide. A few dozen stars are enough to rank new stories by closeness to what you starred.

Later, Claude Haiku can add a judgement per story (“is this a real development or commentary?”) for about a cent a day at 30 stories.

4.7 More sources

Kind of source	What it adds	Effort
Feeds in bulk	Most news sites, think tanks, Substack newsletters and government press pages have a feed. Paste fifty addresses at once.	Small. Use *Quick detect*, which is free; a Claude analysis cost you 11 to 38 cents per source so far.
Your older scrapers	StratCom, CCDCOE, WODC and NATO STO from your mi-scrape project, with full archives.	One built scraper each, as for Clingendael.
GDELT per topic	For each topic you follow, the most-covered articles worldwide of the last day, then read on their own sites. Finds sources you did not know.	Medium. Free; one request per five seconds.
YouTube and podcasts	See 4.8.	Small to medium.
Research and official papers	arXiv, OpenAlex, EU and parliament documents.	Medium; each has an open interface.
Bluesky, Telegram, Mastodon	Fast, short posts from analysts and officials.	Medium. I would wait until the topic pages exist; short posts add noise.

4.8 YouTube and podcasts

There are several ways to get at YouTube, and they differ in what YouTube allows.

Route	What you get	Rules and cost	Verdict
Channel feed	The 15 newest videos of a channel: title, date, description, thumbnail, views. Tested on one channel; the descriptions were 1,000 to 3,000 characters, enough to match a video to a topic.	Free, no key. It is a feed made for feed readers, like Reddit's. YouTube's robots.txt does list it as closed to crawlers.	**Use** to follow channels.
YouTube Data API	A channel's whole archive, video length, and search by keyword ("videos about the Hormuz blockade").	Official and free with a Google key. 10,000 units a day and 100 searches a day. Stored details must be refreshed or deleted every 30 days.	**Use** for archives and topic search.
Embedded player	The video plays on your topic page.	Official, free.	**Use.**
Gemini reads the video	Google's own AI is given the video's address and returns a summary with time stamps.	Official, public videos only. In preview and free of charge now; the free tier allows 8 hours of video a day, and Google may use free-tier content to improve its products. At Google's listed token prices a 20-minute video would cost about 4 to 9 cents.	**Use later**, for summaries. Not tested: it needs a Google AI key.
Unofficial transcript tools	The captions as text.	Free, but outside YouTube's terms, which forbid automated access, and outside its robots.txt. The official API only gives captions to someone who may edit the video.	**Leave out.** The platform's own rule is to stay inside robots.txt.

Route	What you get	Rules and cost	Verdict
Download the audio and transcribe it	A transcript made on your own graphics card.	Downloading from YouTube is against its terms.	**Not for YouTube. Yes for podcasts**, whose feeds hand out the audio file.

> **My pick.** Start with channels as a new kind of source and a video card with the player. Videos then appear on topic pages by meaning, from their title and description. Add topic search with the Data API next. Add Gemini summaries last, as a switch per channel. Many channels worth following also publish as podcasts; your graphics card (RTX 5060 Ti, 16 GB) can transcribe those for free, which I have not tested here.

5. What it costs to run

These are totals for a concrete case, not prices per item: **100 sources, 300 new items a day (9,000 a month), 20 topics you follow, about 3 new videos a day.** Prices are those published in October 2026: Claude Haiku 5.5 $0.10 in and $0.50 out per million tokens, Sonnet 5.5 twenty times that, Opus 5.5 forty times.

Step	Model	Per month	Basis
Embeddings, stories, grouping, trends, scores	None (local)	**$0**	Measured on your computer.
English summary and tags for every item	Haiku	**$3–6**	The platform's estimate: $0.0004 per news article, $0.003 for the longest reports. English text usually costs about half the estimate.
Headline and summary per story (about 30 a day)	Haiku	**$0.30**	My estimate from token counts.
“What is new” on 20 topic pages, daily	Haiku Sonnet Opus	**$0.30** **$6** **$12**	My estimate: 3,000 tokens in and 400 out per update.
“The long run” on 20 topic pages, monthly	Haiku Opus	**$0.10** **$4**	My estimate: 30,000 tokens in, about 1,500 out.
Video summaries (about 90 a month)	Gemini	**$0** now	Free during Google's preview. At listed prices $3 to $8 for 20-minute videos.
Translation, if a third of the items need it	Haiku	**$3**	Measured earlier: about $1 per thousand news articles, more for long papers.

Together: about $4 to $7 a month with Haiku everywhere, about $3 more with translation, and $8 to $16 more if Sonnet or Opus writes the topic pages. I would let Haiku do the per-item work and try Sonnet for the topic pages, because that is the text you actually read. For your library as it is today the same steps cost well under a dollar in total.

Only the summary and translation rows rest on measurements; the others are my estimates and should be measured on the first real run. Every new Claude step should get what translation has now: a spending limit per run and its price shown

before it runs. Anthropic's batch service halves the price of work that can wait (most batches finish within an hour); at these amounts it is not worth the extra complexity yet.

6. Other things that can be done

- **A morning digest.** One page, or an e-mail, with the top stories and what changed in your topics.
- **Alerts.** A notice when a topic spikes or a name you watch appears.
- **Ask your library.** A question box that answers from your stored items, with source numbers.
- **Pages for people, organisations and places.** Who appears in which stories, and with whom.
- **Compare sources.** How think tanks, news and Reddit describe the same story, and where they disagree.
- **Browse the map.** Figure 2 as a page you can zoom and click.
- **Stories on a world map.**
- **Story mode.** One story per screen, next with a key: the closest thing to YouTube Shorts.
- **"What did I miss?"** Everything since your last visit, per topic.
- **Stars, notes, export** to Markdown for a notes app.
- **Listen.** The daily digest read aloud.
- **Run it on your home server,** so that it collects while your PC is off. It needs a login before it is reachable from your phone.
- **A local AI model for summaries.** Possible on your graphics card, but Haiku is better and costs a few dollars a month.

7. Recommended order

Step	What is built	What you can do afterwards	Size and running cost
Today	Nothing. Set update intervals; switch on *AI summary and tags*.	The library grows evenly; every item has an English summary.	$0.32 once for the stored items (Haiku).
1	Embeddings; search by meaning; "more like this".	Ask in plain English, find Dutch and German items.	Small. Free to run.
2	Stories; story cards on the home page; a first importance score.	See events instead of duplicates; sort by coverage.	Medium. Cents a month.
3	The new topic page; suggested topics; "up next" and keys to flip.	The page in Figure 1. This is the centre of what you described.	Large. $0.30 to $12 a month by model.
4	YouTube channels and podcasts as sources; video card; later Gemini summaries.	Videos next to the articles on each topic page.	Small, then medium. Free now.
5	Trends page; weekly share per topic; the world's curve from GDELT.	See what is spiking and what is building.	Medium. Free.
Later	Learning from your stars; digest; alerts; ask your library.		Each small to medium.

Step 4 does not depend on steps 2 and 3, so it can move forward if you want videos sooner. Adding sources fits anywhere. I would not start with step 3: without embeddings and stories the new page would have nothing better to show than the current one.

8. Limits and risks

- **Summaries can be wrong.** Claude can misread or merge two stories. Keeping a source number on every sentence, as the briefing does, is the protection; the page should never show a claim without one.
- **Windows blocked one library during the test.** Smart App Control is switched on on your PC and refused to load scikit-learn, a common grouping toolkit. numpy, scipy and onnxruntime loaded normally, and they are all this plan needs. I did not change the Windows setting. A heavier toolkit such as PyTorch may be blocked as well; I did not try.
- **Automatic groups are rough.** Three of the 28 groups in the test were not subjects, and a few more were loose. Treat them as suggestions.
- **Trends need history and a steady schedule** (section 4.5).
- **YouTube.** The Data API asks that stored video details are refreshed or deleted every 30 days and limits what may be computed from them. A monthly refresh covers the first; trend counts should treat videos as a small extra, not as their base.
- **This is a personal library.** Storing full texts for your own reading is one thing; publishing them or their summaries is another, and the platform is not built for that.
- **The PC has to be on** for scheduled scraping, unless the platform moves to your server.

9. How I checked this

Tested on your computer (free, on a read-only copy of your items in a temporary folder that I removed afterwards): embedding speed with two models, search by meaning, stories, groups, the map, search speed at 100,000 items, one YouTube channel feed, one GDELT query, and what Windows allows to load.

Taken from the official pages on 11 October 2026: YouTube's robots.txt and terms of service; the YouTube Data API pages on caption downloads, quota and developer policies; Google's Gemini pages on video understanding and pricing; Anthropic's pages on embeddings, batch processing and pricing; Voyage AI pricing; the GDELT DOC 2.0 API.

Not tested: grouping on English AI summaries (it would have cost about $0.32), Gemini's video summaries (no Google key), speech-to-text for podcasts, and the embedding models on your graphics card. All cost figures in section 5 other than the summary and translation rows are estimates from token counts.